# Sum-of-Sources-এর বাইরে আর কী আছে?

## ARC White-Box Estimation Challenge 2026 - Phase 2

Alternative formulations for 1024 x 16 bias-free ReLU MLPs
Target lens: raw final-layer MSE ~2e-8 with compute <= 0.16 B

**এক বাক্যের উত্তর:** না - sum-of-sources একমাত্র formulation নয়। Public evidence অনেক নির্দিষ্ট implementation-কে ব্যর্থ দেখিয়েছে, কিন্তু অন্তত 10+ mathematically distinct state representations এখনো logically open। সবচেয়ে আকর্ষণীয় চারটি হলো: (i) quenched TAP/AMP/cavity, (ii) gate-field + amplitude model, (iii) operator / kernel-mean embedding, এবং (iv) input-function TT/PCE compression।

### Research scope

Excluded on purpose: plain Monte Carlo/QMC, ordinary multilevel MC, standard K2/K3/K4 source-bank propagation, finite-width 1/n diagrammatics, ensemble-only DMFT/replica, direct learned weights->means models, and the already-tested full kappa4-memory chain.

# 1. Executive summary

- Strong conclusion: “সব alternative মরে গেছে” বলা evidence-এর চেয়ে বেশি শক্ত। Public WHest logs মূলত specific implementations মেরেছে; family-level impossibility theorem খুব কম।
- Highest-upside non-source-bank route: TAP/AMP/Expectation-Propagation style scalar messages on the actual weight matrices, with Onsager/cavity corrections. এটি ensemble DMFT নয়; algorithm actual W-তে চলে।
- Most mechanistically targeted route: binary gate field (dichotomized-Gaussian / pairwise max-entropy surrogate) + positive-amplitude marks. ReLU error-এর মূল sign-switching structure সরাসরি model করে।
- Most “different coordinate system” route: approximate the whole input-to-output function in a compressed tensor/spectral basis (TT-PCE / functional tensor train), তারপর expectation degree-0 coefficient থেকে পড়া।
- Most exact-but-risky route: activation-region / Gaussian-polytope integration. Exact in principle because ReLU network is piecewise affine; 1024-D region count is the barrier।
- Borderline but important: saddlepoint/CGF propagation finite cumulant truncation নয়; এটি many cumulants-কে non-polynomialভাবে resum করতে পারে। কিন্তু dependence model না থাকলে এটি আবার ব্যর্থ হবে।
- No public challenge implementation found in this search for: forward AMP/TAP, gate-field DG, TT-PCE/TT-cross, sparse characteristic-function state, kernel-mean embedding/Koopman propagation, or saddlepoint-CGF. “Not found” মানে “কেউ করেনি” নয়; public evidence পাওয়া যায়নি।

সবচেয়ে গুরুত্বপূর্ণ **evidence rule:** একটি 1D conditional-mixture failure adaptive full GMM-কে মারে না; input active-subspace R^2 ~1e-3 TT-PCE rank-structure-কে মারে না; Stein control-variate failure Stein-kernel state-কে মারে না; local-linearized Gaussian-copula failure সব copula closure-কে মারে না।

# 2. Problem geometry and the budget filter

**Official Phase-2 geometry:** width n=1024, depth L=16, zero biases, He-Gaussian weights, x~N(0,I), and per-network budget B=2^41 FLOPs. The official metric uses final-layer MSE times max(0.1, C/B).

এই **report-এর engineering lens:** user target C/B <= 0.16. অর্থাৎ এমন method দরকার যা source-bank-এর O(number_of_sources x n^3) growth এড়িয়ে O(n^2), low-single-digit O(n^3), অথবা strongly compressed function-space work-এ থাকে।

**Decision criterion:** method family “interesting” হলেই চলবে না। প্রতিটি branch-এর জন্য cheap premise test থাকতে হবে, যাতে full 1024x16 port করার আগেই rank explosion / component explosion / closure bias ধরা যায়।

# 3. Public failures: কী সত্যিই বন্ধ, আর কী বন্ধ নয়

| Public result | যা সত্যিই বলা যায় | যা বলা যায় না |
|---|---|---|
| 1D conditional Gaussian mixtures failed; score worsened | fixed 1D/2D conditioning + repeated covariance propagation has poor economics | all adaptive mixture propagation is dead |

| Public result | যা সত্যিই বলা যায় | যা বলা যায় না |
|---|---|---|
| Gaussian scale-mixture closure: ~2.33e-6 raw in 504aldo probe | one scalar/scale latent does not explain the relevant non-Gaussianity | multi-component directional GMM split/merge is dead |
| input active-subspace linear R^2 ~1e-3 in barnobarno probe | a few global input directions do not linearly explain the final residual | tensor-train separability or local/nonlinear subspaces are dead |
| shallow Stein controls gave only small variance reduction | those exact-zero-mean input controls do not repay sampling noise | Stein kernel / zero-bias distributional closure is dead |
| local-linearized copula paper is structurally bad at zero bias | linearization at x=0 sits on every first-layer ReLU kink | all copula models are dead |
| source compression / Tucker / per-source rank ladders hit cliffs | leg/source-based K3 compression has a serious rank wall | all representations of the distribution have the same wall |

## 4. Formulation map: 14 alternatives beyond ordinary sum-of-sources

| # / family | State | Distinctness | Budget plausibility | Research priority |
|---|---|---|---|---|
| 1 TAP/AMP/cavity | scalar/vector messages + Onsager terms | Genuine | Very strong if derivable | A |
| 2 Gate-field DG/Ising + marks | gate probabilities/correlations + positive amplitudes | Genuine | O(n^2)-ish target | A |
| 3 Koopman/KME/Galerkin | finite dictionary of observables / mean embedding | Genuine | K small => strong | A- |
| 4 TT-PCE / input chaos | compressed function coefficients in input coordinates | Genuine axis change | rank-dependent | A- |
| 5 Functional TT / TT-cross integration | black-box function tensor cores | Genuine | rank-dependent | B+ |
| 6 EP / polyhedral gate integration | Gaussian mass over gate inequalities | Genuine | few sweeps / selected gates | B+ |
| 7 Characteristic-function state | sparse Fourier/CF evaluations | Genuine | frequency-rank dependent | B |
| 8 Copula + rich marginals | marginal CDFs + dependence copula | Genuine | O(n^2)-O(n^3) | B |
| 9 Adaptive structured GMM | few split/merge components | Genuine | only if M small | B |

| # / family | State | Distinctness | Budget plausibility | Research priority |
|---|---|---|---|---|
| 10 Transport-map closure | compressed map from Gaussian base to activations | Genuine | rank/order dependent | B- |
| 11 Quenched random-matrix DE | resolvents/local laws + actual-W corrections | Genuine if anisotropic | few n^3 ops possible | A-/B+ |
| 12 Saddlepoint / CGF | log-MGF values; tail via saddlepoint | Borderline: all-cumulant resummation | cheap if dependence solved | A- hybrid |
| 13 Stein-kernel / zero-bias | Stein discrepancy / transformed law | Borderline | O(n^2) target | B |
| 14 Angular harmonic / heat-semigroup | spectral/semigroup representation | Borderline with chaos/PCE | compression needed | C+ |

## 1. Quenched TAP / AMP / cavity message passing

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| শ্রেণি | GENUINELY DISTINCT - HIGH UPSIDE |
| **মূল representation** | Per-neuron (or small vector) messages: cavity mean/variance, susceptibility, and Onsager reaction terms; no per-birth K3 source bank. |
| **কেন sum-of-sources নয়** | Forward distribution is represented by local messages on the dense random linear maps, not by a global third-order tensor. TAP/AMP can run on the actual W, unlike an ensemble-only DMFT scalar recursion. |
| **Phase-2 fit** | Dense i.i.d. Gaussian matrices are exactly the regime where AMP-style state evolution and Onsager corrections are strongest. If a forward-marginal analogue can be derived, each iteration can be O(L n^2) rather than O(number_of_sources n^3). |
| প্রধান ঝুঁকি | Existing ML-VAMP/GAMP theory is mostly inverse inference with observations. With no observations, a naive adaptation may collapse to ordinary Gaussian mean-field and lose the finite-n quenched correction we need. |
| **প্রথম falsifier** | Derive a one-layer cavity/EP update for z=W^T h with h distributed from a teacher-forced previous layer. On 20 MLPs compare one-step final-mean error against K2 and K3. Kill if Onsager/cavity terms give <30% residual reduction or if >4-8 sweeps are needed. |

**Evidence note:** ML-VAMP and ML-BiGAMP show that multilayer dense random nonlinear models admit scalar/state-evolution message passing in large-system limits; no WHest-specific forward estimator was found.

## 2. Gate-field model: dichotomized Gaussian / max-entropy gates + amplitude marks

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| শ্রেণি | GENUINELY DISTINCT - HIGH UPSIDE |
| **মূল representation** | Separate h_i = g_i a_i with binary gate g_i=1[z_i>0] and positive amplitude a_i. Track gate rates, pair correlations, and marked conditional amplitudes. |

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| কেন **sum-of-sources** নয় | The state is a binary switching field plus amplitudes, directly modeling ReLU sign structure instead of polynomial cumulants of h. |
| **Phase-2 fit** | At layer 1, gates are exactly thresholded jointly Gaussian. A DG surrogate naturally matches binary rates/covariances; a marked version could preserve E[h_i], E[h_i h_j], and gate-amplitude coupling with O(n^2) state. |
| প্রধান ঝুঁকি | Pairwise gate statistics may be insufficient; partition functions / high-order gate clusters can become hard. A simplistic marked-DG can collapse to covariance propagation. |
| প্রথম **falsifier** | Use offline MC/high-moment diagnostics on 128-neuron subsets at layers 4/8/12. Fit DG from gate means/pairs, then test triple gate probabilities and one-step E[ReLU(W^T h)]. Kill if marked-DG fails to beat Gaussian/covariance closure by >=20% on one-step residual. |

**Evidence note:** DG models are established for high-dimensional correlated binary fields. Public WHest searches found no gate-field implementation.

## 3. Finite Koopman / transfer-operator or kernel-mean embedding (KME)

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| শ্রেণি | GENUINELY DISTINCT |
| মূল **representation** | Represent the distribution only through expectations of a chosen observable dictionary psi_1,...,psi_K (or RKHS mean embedding); propagate observables under each deterministic layer. |
| কেন **sum-of-sources** নয় | This propagates linear functionals of the law in feature space rather than raw moments/cumulants or samples. A basis can include gate-sensitive nonlinear observables. |
| **Phase-2 fit** | If K is tens to a few hundreds, applying W to basis summaries can be O(K n^2). A sparse conditional-mean/Koopman operator gives a natural place to retain exactly the observables that predict E[ReLU]. |
| প্রধান ঝুঁকি | Closure is the whole problem: generic nonlinear maps need an infinite observable space. A poor dictionary becomes another low-order moment method; an over-rich dictionary explodes. |
| প্রথম **falsifier** | Start with 32-128 analytic observables (CDF/gate, \|z\|, z_+, z_+^2, selected pair gates) and fit/projection using only public training MLPs or analytic quadrature. On held-out networks test whether the next-layer mean lies in the span with >90% residual reduction. Kill if K must grow with n. |

**Evidence note:** KME/CME uncertainty propagation and Koopman lifting are established in nonlinear dynamical systems, including sparse operators.

## 4. Input-space polynomial chaos in TT/CP form (TT-PCE)

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| শ্রেণি | GENUINELY DISTINCT AXIS |
| মূল **representation** | Approximate each target neuron as a function f_j(x) of the original Gaussian input using Hermite/PCE coefficients compressed in tensor-train or canonical low-rank format. |
| কেন **sum-of-sources** নয় | K3 source propagation compresses an output cumulant tensor across neurons/sources; TT-PCE compresses the input-coordinate dependence of the deterministic network function. Expectation is |

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| | simply the degree-0 coefficient. |
| **Phase-2 fit** | The first layer is a ridge function, which is extremely structured in input space. Tensorized PCE can beat the combinatorial basis size when TT ranks stay small. |
| প্রধান ঝুঁকি | Dense random W may destroy separability quickly. The chaos spectrum in public WHest probes has a long high-order tail, so low polynomial degree alone is not enough; success requires low tensor rank despite high degree. |
| প্রথম **falsifier** | On n=64/128, depth 16, run TT-cross/TT-PCE for 8-16 chosen outputs. Measure TT rank vs depth and integral error. Kill if median TT rank >32-64 by layer 4 or projected model evaluations exceed the 0.16B budget. |

**Evidence note:** Low-rank PCE and tensor-train uncertainty quantification are established high-dimensional tools; this exact WHest compression axis appears publicly untested.

## 5. Functional tensor train / TT-cross direct Gaussian integration

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| শ্রেণি | GENUINELY DISTINCT |
| মূল **representation** | Treat f_j(x) itself as a tensorized black-box function on product Gaussian quadrature grids; recover low-rank TT cores adaptively, then integrate cores exactly. |
| কেন **sum-of-sources** নয় | No layerwise law is propagated at all. The algorithm approximates the final deterministic function and integrates it. |
| **Phase-2 fit** | TT-cross can require O(d q r^2) function probes rather than q^d when ranks r are small; Gaussian measure is separable. |
| প্রধান ঝুঁকি | Each function probe is a full 16-layer forward pass. Dense random mixing can make TT ranks huge; then it degenerates into expensive cubature. |
| প্রথম **falsifier** | Small-width rank-growth probe first. Compare QTT/TT-cross rank for final neurons across depth 2,4,8,16. Kill immediately if ranks scale roughly linearly with width or if cross evaluations exceed matched MC count with no large accuracy gain. |

**Evidence note:** Tensor-train integration and tensor neural-network quadrature have polynomial-in-dimension complexity only under low-rank structure.

## 6. Activation-region / Gaussian-polytope integration

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| শ্রেণি | GENUINELY DISTINCT - EXACT IN PRINCIPLE |
| মূল **representation** | A ReLU network partitions input space into polyhedra. On each activation region, every neuron is affine; expectation is a sum of truncated-Gaussian affine moments over regions. |
| কেন **sum-of-sources** নয় | The state is geometric: activation patterns and Gaussian mass of polytopes, not cumulants. |
| **Phase-2 fit** | For shallow/small PL networks there are analytic moment formulas. A probability-mass beam could in principle keep only regions carrying nearly all Gaussian mass. |

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| প্রধান ঝুঁকি | At 1024x16 the region count is the obvious combinatorial wall. Zero bias places the Gaussian center on many hyperplanes, making mass spread especially dangerous. |
| প্রথম **falsifier** | Estimate activation-pattern mass concentration from offline samples. If top 256/1024 patterns capture only a tiny fraction of mass, kill full-region beam. A more local variant can condition only on the last 1-2 layers and use EP for their gate polyhedra. |

**Evidence note:** Exact/analytic Gaussian moments of piecewise-linear networks exist for limited architectures; general ReLU region geometry is polyhedral and can grow rapidly.

## 7. Expectation Propagation over gate inequalities / polyhedral cavity

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| শ্রেণি | GENUINELY DISTINCT |
| মূল **representation** | Represent active/inactive ReLU constraints as truncation factors; iteratively replace them with Gaussian site factors using cavity distributions and moment matching. |
| কেন **sum-of-sources** নয় | EP works with approximate factors/partition functions and truncated Gaussian geometry rather than carrying third-order source tensors. |
| **Phase-2 fit** | Rank-one site updates can be cheap. A block-EP restricted to influential gates could add non-Gaussian gate information on top of a Gaussian backbone without explicit region enumeration. |
| প্রধান ঝুঁকি | EP can be arbitrarily wrong for general polyhedra; 1024 gates per layer can require too many site updates. ReLU is not just an indicator - active amplitude must also be handled. |
| প্রথম **falsifier** | At one layer with teacher-forced Gaussian input, compare EP-predicted first/second moments after 32,64,128 selected gate factors to exact MC. Kill if error does not fall monotonically or if full sweep count makes cost >~5u/layer. |

**Evidence note:** Cunningham-Hennig-Lacoste-Julien show EP is accurate for many Gaussian regions but can fail badly on arbitrary polyhedra - so it needs aggressive falsification, not blind trust.

## 8. Characteristic-function / sparse Fourier propagation

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| শ্রেণি | GENUINELY DISTINCT |
| মূল **representation** | Track phi(t)=E exp(i t^T h) at selected frequencies, or low-rank marginal/joint characteristic-function features. |
| কেন **sum-of-sources** নয় | Characteristic functions describe the law in frequency space and do not require finite moments. Linear transforms are exact via phi_{Wh}(t)=phi_h(W^T t). |
| **Phase-2 fit** | The linear step is ideal. If only a sparse set of frequencies around directions relevant to the next W rows is needed, a small adaptive frequency bank could capture non-Gaussian shape near the ReLU threshold. |
| প্রধান ঝুঁকি | Coordinatewise ReLU couples frequencies and high-dimensional inversion is expensive. Full multivariate CF is impossible at n=1024; sparse frequency closure may lose dependence. |
| প্রথম **falsifier** | Teacher-force one layer and retain q=16,32,64 adaptive frequencies per target. Reconstruct E[z_+] |

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| | from the CF and compare with K3. Kill if q grows rapidly with depth or if joint-frequency rank is high. |

**Evidence note:** Pilipovsky et al. explicitly propagate distributions of ReLU networks through characteristic functions for probabilistic verification.

## 9. Gaussian copula + rich nonparametric marginals

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| শ্রেণি | GENUINELY DISTINCT IF MARGINALS ARE RICH |
| মূল **representation** | Track each neuron marginal by quantile/PDF knots and dependence via a Gaussian or low-rank copula correlation matrix. |
| কেন **sum-of-sources** নয় | Non-Gaussianity is stored in one-dimensional marginals; dependence is separated into a copula rather than encoded in joint cumulant tensors. |
| **Phase-2 fit** | Marginal ReLU is exact (truncate/atom at zero), and O(q n) marginal state is cheap. Pair dependence can reuse an O(n^2) correlation backbone. |
| প্রধান ঝুঁকি | A local-linearized copula model is poor for zero-bias ReLU because x=0 lies on the kink. Weighted sums under a copula still need an efficient transform; Gaussian copula may miss tail/gate dependence. |
| প্রথম **falsifier** | Do not test the published local-linearization directly. Instead teacher-force true marginals, fit only the copula, and test next-layer positive-part mean. If even oracle marginals + copula cannot beat K2/K3, kill the family. |

**Evidence note:** 2026 Diamzon-Venturi derives PDF/moment propagation with Gaussian-copula surrogates, but 504aldo correctly noted their linearization assumptions are unfavorable for this challenge. That does not falsify an oracle-marginal copula test.

## 10. Adaptive structured Gaussian mixture split/merge

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| শ্রেণি | GENUINELY DISTINCT |
| মূল **representation** | Approximate the activation distribution by M Gaussian components, splitting components where ReLU nonlinearity causes largest Wasserstein distortion and merging redundant ones. |
| কেন **sum-of-sources** নয় | The state is a multimodal density, not a polynomial moment hierarchy. |
| **Phase-2 fit** | WGMprop-style adaptive split/merge is explicitly designed for nonlinear neural-network uncertainty. A structured covariance (diag + rank-r) could keep each component much cheaper than full covariance. |
| প্রধান ঝুঁকি | 1024 simultaneous gates can force component explosion. Public WHest failures show 1D conditional mixtures and a Gaussian scale-mixture are insufficient, but not that adaptive directional GMM is impossible. |
| প্রথম **falsifier** | Implement only on width 128/256 first. Track number of components needed for one-step and rolled mean error. Kill if M>16-32 by layer 4 or if component count doubles repeatedly. |

**Evidence note:** Monchot et al. (ICML 2023) and earlier work use adaptive GMM split/merge through neural networks; public WHest tests covered much narrower mixture classes.

## 11. Quenched random-matrix deterministic equivalents + anisotropic correction

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| শ্রেণি | GENUINELY DISTINCT IF SAMPLE-SPECIFIC |
| মূল **representation** | Use actual-W Gram/resolvent/local-law observables to approximate the finite-width distribution; retain a small set of anisotropic diagonal or leave-one-out corrections. |
| কেন **sum-of-sources** নয় | Unlike ensemble DMFT, quenched/local-law methods can depend on the realized matrices. Unlike K3 source banks, the state can be a few resolvents, traces, diagonals, or susceptibilities. |
| **Phase-2 fit** | Random He matrices are exactly random-matrix objects. Deterministic-equivalent and neural-matrix-law literature suggests many bulk quantities self-average; the unexplained residual may live in a small anisotropic correction. |
| প্রধান ঝুঁকি | Per-neuron 2e-8 accuracy may depend on fluctuations not captured by bulk spectra. A pure deterministic equivalent would only give a typical/ensemble result and fail the task. |
| প্রথম **falsifier** | Build a residual regression/falsifier using only actual-W observables: row norms, diagonal powers of Gram matrices, trace moments, approximate resolvent diagonals, leverage/susceptibility. Cross-validate across MLPs. If these explain <20-30% of V29 residual, scalar DE route is unlikely. |

**Evidence note:** Tensor Programs III proves asymptotic freeness; deep random-feature work derives deterministic equivalents. The missing piece is a quenched, per-neuron finite-n correction.

## 12. Saddlepoint / cumulant-generating-function (CGF) propagation

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| শ্রেণি | BORDERLINE: NOT FINITE-ORDER CUMULANT TRUNCATION |
| মূল **representation** | Track K(t)=log E[e^{tZ}] at selected t and use a saddlepoint (e.g. Lugannani-Rice / Daniels) to recover tail probabilities or E[Z_+]. |
| কেন **sum-of-sources** নয় | It does not store kappa3 source tensors and does not truncate after kappa3/kappa4; a saddlepoint can effectively resum all cumulants encoded by K(t). |
| **Phase-2 fit** | For a weighted sum of independent/weakly dependent terms, CGFs add. If TAP/cavity can produce near-independent effective channels, CGF+saddlepoint becomes especially attractive and cheap. |
| প্রধান ঝুঁকি | The layer activations are dependent; naive product-CGF ignores the exact thing K3 is fixing. Without a dependence/cluster correction this may revert to Gaussian-quality error. |
| প্রথম **falsifier** | Teacher-force h samples, compute “oracle marginal CGF + independence” prediction for W^T h, then add pair/Bethe corrections. If oracle-marginal independence is already far above 1e-7 one-step MSE, do not port. |

**Evidence note:** Classical saddlepoint approximations are much more uniform in tails than Edgeworth expansions for sums, but the independence assumption is the key obstacle here.

## 13. Stein-kernel / zero-bias distributional closure

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| শ্রেণি | BORDERLINE BUT DIFFERENT FROM STEIN CONTROL VARIATES |
| মূল **representation** | Track a Stein kernel, zero-bias transform, or small discrepancy features that directly correct Gaussian integration-by-parts identities for each preactivation. |
| কেন **sum-of-sources** নয় | A Stein control variate is a sampling trick; a Stein-kernel state is a deterministic description of how the law deviates from Gaussian and can target E[ReLU(Z)] directly. |
| **Phase-2 fit** | Stein theory already gives quantitative Gaussian approximation bounds for deep random Gaussian networks. A scalar/pairwise Stein discrepancy might capture the ReLU mean error with much less state than full K3. |
| প্রধান ঝুঁকি | No known closed recurrent Stein-kernel update for dependent ReLU layers. It may secretly require the same high-order information as cumulants. |
| প্রথম **falsifier** | Offline teacher-forced test: regress the exact ReLU-mean residual against candidate Stein observables (density-at-zero proxy, zero-bias displacement, pairwise Stein matrix summaries). Kill if out-of-sample R^2 is low or correction is not stable across depth. |

**Evidence note:** Public WHest Stein-CV failures do not test this deterministic-state idea. Stein-method literature supplies approximation identities/bounds, not a ready competition estimator.

## 14. Angular harmonics / heat-semigroup / spectral smoothing

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| শ্রেণি | BORDERLINE SPECTRAL FAMILY |
| মূল **representation** | After exact radial separation, represent the angular function on the sphere via low-degree spherical harmonics/Gegenbauer ridge components; equivalently view Gaussian expectation as a heat-semigroup smoothing of the CPWL function. |
| কেন **sum-of-sources** নয় | This works in function/spectral space rather than neuron cumulant space. The mean is the degree-0 mode after angular propagation. |
| **Phase-2 fit** | Positive homogeneity is unusually favorable: radius separates exactly. ReLU ridge functions have structured spherical harmonic expansions, so a compressed zonal representation is conceivable. |
| প্রধান ঝুঁকি | The number of harmonics in d=1024 explodes with degree; public chaos-spectrum measurements show a heavy high-order tail. Without tensor compression this is dead, and with low-degree truncation it may become ordinary Hermite moment closure. |
| প্রথম **falsifier** | On small widths measure angular spectral energy and tensor rank by degree. Kill if degree>6-8 carries material mean correction or coefficient rank explodes. Keep only if a Gram/zonal representation avoids explicit harmonic enumeration. |

**Evidence note:** Useful mainly as a basis-design idea for TT/operator methods, not as a standalone explicit harmonic expansion.

## 5. The strongest new combinations (not individual methods)

**Combination A - TAP/AMP + saddlepoint CGF** Use cavity/Onsager to turn each dense weighted sum into an approximately decoupled scalar channel, then use a saddlepoint rather than a Gaussian/Edgeworth truncation for the scalar channel. যদি কাজ করে, source bank-এর dependence correction message-passing-এ যায়, আর non-Gaussian marginal shape CGF-এ যায়।

**Combination B - Gate field + marked amplitudes + EP** DG/Ising-style gate correlations capture sign structure; EP handles selected high-influence gate constraints; conditional positive amplitudes carry magnitude. This attacks the ReLU kink directly rather than reconstructing a global K3 tensor.

**Combination C - Angular radial separation + TT-PCE/TT-cross** Radius is analytically integrated once (already known). The remaining sphere/input-direction function is compressed in TT/PCE. This avoids wasting tensor rank on a nuisance radial dimension and is much more interesting than plain low-degree PCE.

**Combination D - Quenched RMT observables as an operator dictionary** Use resolvent/Gram/leverage/susceptibility features from actual W as the low-dimensional Koopman/KME dictionary. This is a principled way to inject network-specific finite-width information without source history.

## 6. What I would test first: a five-gate research ladder

**Gate 1 - TAP/AMP feasibility (highest information value)**: Before full derivation, test whether an Onsager/cavity correction computed from actual W and teacher-forced previous-layer statistics predicts the one-step residual direction. Use 20 public MLPs, layers 4/8/12/15. Promotion: >=30% reduction in one-step mean MSE on held-out MLPs with O(n^2) or <=2u equivalent extra work.

**Gate 2 - Marked gate-field oracle**: Give the method oracle gate means/pair correlations from MC and test whether DG + conditional amplitude reconstructs the next-layer mean. This separates “representation insufficient” from “parameter estimation insufficient.” Promotion: clear gain over K2/K3 one-step closure; reject if oracle pairwise gate state is still weak.

**Gate 3 - TT rank-growth probe**: At n=64/128, approximate 8 final neurons with TT-cross/TT-PCE and record TT rank by depth. Promotion: ranks stay <=32-ish and integral error decays faster than matched forward evaluations. Reject on early rank explosion.

**Gate 4 - CGF/saddlepoint oracle-dependence test**: From teacher samples compute exact univariate marginal CGFs, first ignoring dependence and then adding pairwise/Bethe corrections. If even oracle marginals cannot produce an accurate weighted-sum positive-part mean, the saddlepoint branch is closed.

**Gate 5 - sparse operator dictionary**: Build a 32/64/128-feature deterministic dictionary mixing gate probabilities, marginal transforms, Gram/resolvent diagonals, and angular features. Fit/projection only on train MLPs; evaluate full rollout on untouched MLPs. Promotion only if error reduction survives without K growing with n.

## 7. Which routes could realistically hit <=0.16B?

| Route | Compute shape if successful | Accuracy ceiling risk | Verdict before experiments |
|---|---|---|---|
| TAP/AMP/cavity | O(iter x L n^2) to low n^3 | may collapse to mean-field | Best theoretical gamble |
| Gate-field + marked amplitude | O(L n^2) state; pair updates | pairwise gates may miss clusters | Best mechanism-targeted gamble |
| TT-PCE / TT-cross | O(d q r^2) probes/ops if rank low | rank explosion | Best representation gamble |
| Sparse operator/KME | O(K L n^2) if K small | dictionary closure | Very testable |
| Saddlepoint-CGF hybrid | cheap after decoupling | dependence is hard | Good add-on, not standalone |
| Copula + marginals | O(n^2)-few n^3 | copula bias | Medium |
| Adaptive GMM | M x covariance cost | component explosion | Medium-low |
| Full region/polytope | combinatorial | region explosion | Low standalone |

## 8. Bottom line

**Sum-of-sources "the only possible formulation"** নয়। এটি বর্তমান K3 architecture-এর একটি very effective representation, কিন্তু probability law, gate geometry, observables, Fourier space, input-function space, message space, অথবা quenched random-matrix state - এসব সম্পূর্ণ আলাদা coordinate system-এ একই expectation problem attack করা যায়।

তবে সবচেয়ে বড় **lesson:** alternative family-এর নাম পাওয়া breakthrough নয়। Breakthrough হবে এমন একটি representation পাওয়া যার state dimension depth-এর সাথে source count-এর মতো বাড়ে না এবং তবু fixed-W finite-width correction ধরে। এই criterion-এ TAP/AMP, marked gate-field, sparse operator/KME, এবং TT-function compression সবচেয়ে পরিষ্কার নতুন bets।

বর্তমান **evidence** থেকে আমার **research order:** 1) TAP/AMP one-step derivation/falsifier; 2) marked gate-field oracle; 3) TT rank probe; 4) CGF saddlepoint hybrid; 5) sparse operator dictionary. এগুলো fail করলে তারপর copula/GMM/polyhedral branches-এ যাওয়া যৌক্তিক।

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**Research caveat:** The report separates published facts from proposed WHest adaptations. AMP/TAP forward propagation, marked gate-field, TT-PCE at 1024x16, and quenched DE corrections are research hypotheses here, not measured competition results. “No public implementation found” is not evidence of nonexistence.